Binary Floating-Point Representation

IEEE-754

The IEEE-754 standard defines binary floating-point arithmetic formats used in most modern computer systems. It provides standardized representation for real numbers using scientific notation in binary form.

Binary Floating Point Encoding Rules

- Significand in binary
- Binary representation in base-2
- Normalized form: $1.f \times 2^E$
- Sign encoding: Positive = 0, Negative = 1
- Biased exponent: E' = E + 127

IEEE-754 Single-Precision (32-bit)

Format Structure:

- Sign: 1 bit (MSB)
- Exponent: 8 bits (biased by +127)
- Fraction: 23 bits (implicit leading 1)

TABLE 1. The special cases for binary single-precision floating point.

Sign Bit	E'	Significand	Case
0	0000 0000	000 0000 0000 0000 0000 0000	+ 0
1	0000 0000	000 0000 0000 0000 0000 0000	- 0
0 or 1	0000 0000	Not equal to zero	Denormalized
0	1111 1111	000 0000 0000 0000 0000 0000	+ Infinity
1	1111 1111	000 0000 0000 0000 0000 0000	- Infinity
X	1111 1111	01X XXXX XXXX XXXX XXXX XXXX	sNaN
X	1111 1111	1XX XXXX XXXX XXXX XXXX XXXX	qNaN

Example 1: $1.00_2 * 2^2$

Yes | Significand in binary
Yes | Binary representation in base-2
Yes | Normalized form: 1.f × 2E
0 | Sign encoding: Positive = 0, Negative = 1
129 | Biased exponent: E' = E + 127

S = 0 (positive)
E' = E + 127 (bias) = 129 = 1000 0001
F = 0..0

Binary:

S	E	F
0	10000001	0..0

Grouped:
0100 0000 1000 0..0

Hex:
0x4080_0000

Example 2: 4.00_{10}

No | Significand in binary
No | Binary representation in base-2
No | Normalized form: 1.f × 2E
0 | Sign encoding: Positive = 0, Negative = 1
129 | Biased exponent: E' = E + 127

$4.00_{10} * 10^0$
$= 100.0_2 * 2^0$ (turn significand to binary and make it base-2)
$= 1.000_2 * 2^2$ (normalize)

S = 0 (positive)
E' = E + 127 (bias) = 129 = 1000 0001
F = 0..0

Binary:

S	E	F
0	10000001	0..0

Grouped:
0100 0000 1000 0..0

Hex:
0x4080_0000

Example 3: $1.1101_2 * 2^{135}$

```
Yes  | Significand in binary
Yes  | Binary representation in base-2
Yes  | Normalized form: 1.f × 2E
0    | Sign encoding: Positive = 0, Negative = 1
+inf | Biased exponent: E' = E + 127

exponent is too big, automatically special case (infinity)

S = 0 (positive)
E' = special case = 1111 1111
F = 0..0

Binary:
S         E     F
0  11111111  0..0

Grouped:
0111 1111 1000 0..0

Hex:
0x7F80_0000   special case: +infinity
```

Example 4: $1.1101_2 * 2^{-135}$

```
Yes  | Significand in binary
Yes  | Binary representation in base-2
Yes  | Normalized form: 1.f × 2E
0    | Sign encoding: Positive = 0, Negative = 1
dnml | Biased exponent: E' = E + 127

exponent is too small, automatically special case (denormalized)

1.1101₂ * 2^-135
= 0.0000000011101₂ * 2^-126

S = 0 (positive)
E' = special case = 0000 0000
F = 000 0000 0111 010..0

Binary:
S         E                    F
0  00000000  000 0000 0111 010..0

Grouped:
0000 0000 0000 0000 0111 0100 0..0

Hex:
0x0000_7400   special case: denormalized
```

IEEE-754 Double-Precision (64-bit)

Format Structure:

- Sign: 1 bit (MSB)
- Exponent: 11 bits (biased by +1023)
- Fraction: 52 bits (implicit leading 1)

TABLE 2. The special cases for binary double-precision floating point.

Sign Bit	E'	Significand	Case
0	000 0000 0000	0..0	+ 0
1	000 0000 0000	0..0	- 0
0 or 1	000 0000 0000	Not equal to zero	Denormalized
0	111 1111 1111	0..0	+ Infinity
1	111 1111 1111	0..0	- Infinity
X	111 1111 1111	01X..X	sNaN
X	111 1111 1111	1X..X	qNaN

Example 1: $+1000.0_{10}$

No | Significand in binary
No | Binary representation in base-2
No | Normalized form: 1.f × 2E
0 | Sign encoding: Positive = 0, Negative = 1
129 | Biased exponent: E' = E + 127

$+1000.0_{10} * 2^0$
$= 1111101000.0_2 * 2^0$
$= 1.111101_2 * 2^9$

S = 0 (positive)
E' = E + 1023 (bias) = 1032 = 100 0000 1000
F = 1111 010..0

Binary:

S	E	F
0	100 0000 1000	1111 010..0

Grouped:
0100 0000 1000 1111 0100 0..0

Hex:
0x408F_4000_0000_0000